East West University

Course Code & Name: CSE 303 (Statistics for Data Science)

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Course Section: 8

Academic Session: Spring 2026

Final Project Report

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Project Title:

****Air Quality Index Prediction in Bangladesh (2000–2025): A Comparative Study of Linear Regression with Gradient Descent and Random Forest Regression****

Submission Date & Day: April 20, 2026 (Monday)

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Abstract

This study investigates AQI prediction using a large-scale Bangladesh air quality dataset spanning 2000–2025 with over 1 million hourly observations across 103 cities. The dataset includes pollutant concentrations such as PM2.5, PM10, NO2, SO2, O3, CO, and CO2 along with AQI values computed using US EPA standards.

We compare two machine learning approaches: Linear Regression and Random Forest Regression. Extensive exploratory data analysis (EDA), feature engineering, and temporal analysis were performed. Model performance was evaluated using MAE, MSE, RMSE, and R^2.

Random Forest achieved the best performance (MSE ≈ 159.61, R^2 ≈ 0.94), explaining approximately 94% of the variance in AQI values. In contrast, Linear Regression showed comparatively lower performance (MSE ≈ 650.14, R^2 ≈ 0.77), indicating limited capability in capturing complex relationships in air quality data. Gradient Descent improved parameter optimization and reduced training loss over iterations but did not significantly enhance overall predictive performance compared to the closed-form Linear Regression solution.

1. Introdution

Air pollution is a major environmental and public health challenge in Bangladesh, especially in urban regions. Continuous monitoring and accurate prediction of AQI can support policy-making and early warning systems.

With the availability of high-resolution environmental datasets, machine learning models provide an effective approach to predict AQI based on pollutant concentrations and temporal patterns. This study investigates the performance of Linear Regression and Random Forest Regression on a large-scale AQI dataset covering 25 years.

2. Dataset Description

Bangladesh Air Quality Index (AQI) is collected from "Bangladesh Air Quality Index (AQI) Dataset (2000–2025): Historical Hourly Air Pollution Data Across 103 Cities" which is a large-scale, high-resolution environmental dataset providing historical air pollution measurements across 103 cities in Bangladesh. It is obtained from Mendeley and processed in Kaggle environment.

Dataset Platform: Mendeley Data.
Source link: https://data.mendeley.com/datasets/9j447cynb9/2

2.1 Key Characteristics

- **Records:** 1,048,551 hourly observations.
- **Cities:** 103 across Bangladesh.
- **Features (initial):** PM2.5, PM10, CO, CO_2, NO_2, SO_2, O_3, AQI, timestamps, city metadata, geographic coordinates.
- **Target Variable:** AQI (EPA-based index)

2.2 Description of each features

Feature	Type	Classification	Description
City_Id	Integer	Categorical(Nominal)	City wise unique ID
City_name	Object	Categorical(Nominal)	Name of each cities
Lat	Float	Numerical(Continuous)	Geographic coordinate(latitude)
Lon	Float	Numerical(Continuous)	Geographic coordinate(longitude)
Datetime	Object	Categorical(Ordinal)	Timestamps in ISO 8601 format
PM10	Float	Numerical(Continuous)	Concentrations of air pollutants
PM2.5	Float	Numerical(Continuous)	Concentrations of air pollutants
Carbon monoxide	Float	Numerical(Continuous)	Concentrations of air pollutants
Carbon dioxide	Float	Numerical(Continuous)	Concentrations of air pollutants
Nitrogen dioxide	Float	Numerical(Continuous)	Concentrations of air pollutants
Sulphur dioxide	Float	Numerical(Continuous)	Concentrations of air pollutants
Ozone	Float	Numerical(Continuous)	Concentrations of air pollutants
AQI	Float	Numerical(Continuous)	Composite Air Quality Index

TABLE I: Descriptions of each features.

After downloading the dataset from Mendeley Data, it was uploaded to the Kaggle to perform analysis in a cloud-based environment. Kaggle was chosen due to its computational efficiency and suitability for handling large datasets.

- Kaggle dataset link: https://www.kaggle.com/datasets/sakinofbd/bangladesh-aqi-dataset-20002025

2.3 Data Cleaning

```
city_id                0
city_name              0
lat                    0
lon                    0
datetime               0
pm10                   0
pm2_5                  0
carbon_monoxide        0
carbon_dioxide    774327
nitrogen_dioxide       0
sulphur_dioxide        0
ozone                  0
aqi                  696
dtype: int64
```

Fig 1: Number of Null values in each column

- Removed CO2 column (74% missing values)
- Removed AQI missing rows (696 values)
- Removed city_id and city_name (non-predictive identifiers)

Finally, There are 1047855 Rows remaining in the dataset.

3. Exploratory Data Analysis (EDA)

3.1 Descriptive Statistics

Feature	Mean	Median	Mode	Standard deviation	Variance	IQR
PM10	75.01198	56.1	10.0	67.45101	4549.6389	73.600
PM2.5	50.7021	39.9	10.2	41.8621	1752.4427	50.5999
Carbon Monoxide	323.6237	272.0	177.0	280.4720	78664.5535	296.0
Nitrogen Dioxide	16.7857	10.7	1.5	17.3524	301.1080	20.0
Sulphur Dioxide	9.7886	6.6	2.0	9.3027	86.5410	11.2999
Ozone	73.8014	64.0	55.0	41.0441	1684.6212	49.0
AQI	115.0750	117.0	151.0	52.6279	2769.7007	89.2354

TABLE II: Mean, Median, Mode and distribution statistics for all features.

Here the table above shows the numerical features such as mean, median, mode, standard deviation ,variance and IQR of the data where the majority of air pollutants have right-skewed distributions, with mean values exceeding the median and mode, indicating the presence of extreme pollution. High standard deviation and IQR values across most variables suggest significant variability in pollutant concentrations. AQI also shows a wide spread, reflecting fluctuating air quality conditions across different regions and time periods.

3.2 Distribution of Pollutants

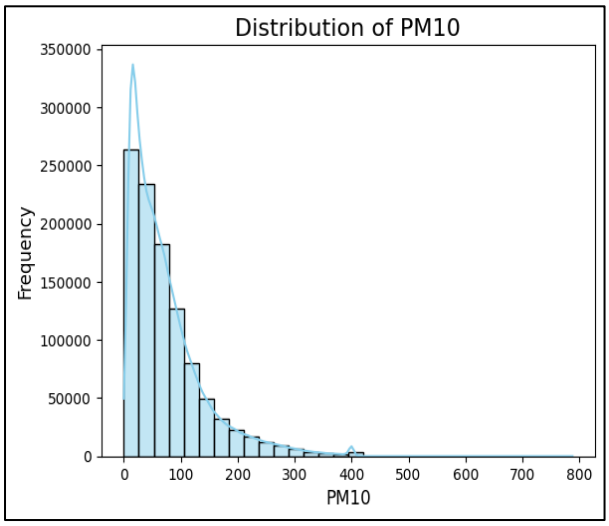


Fig 2: Histogram of PM10.

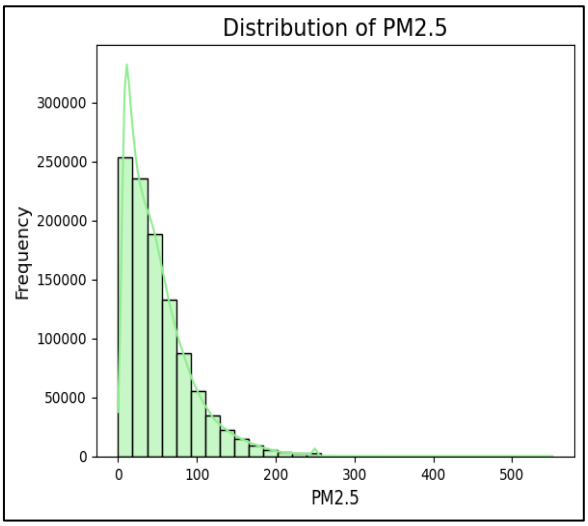


Fig 3: Histogram of PM2.5.

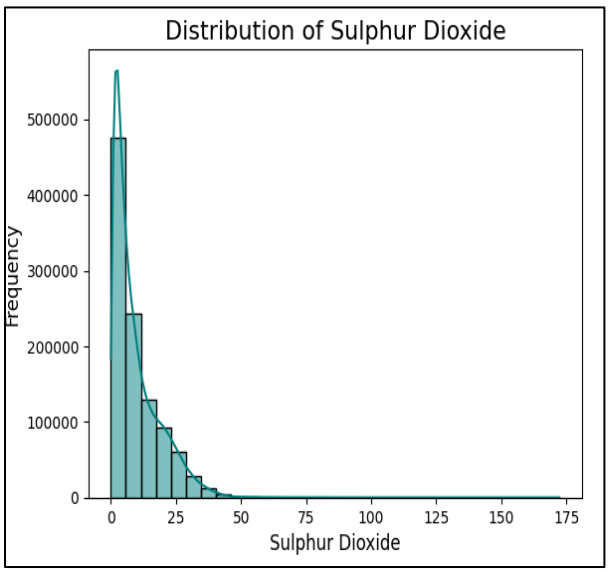


Fig 4: Histogram of SO_2.

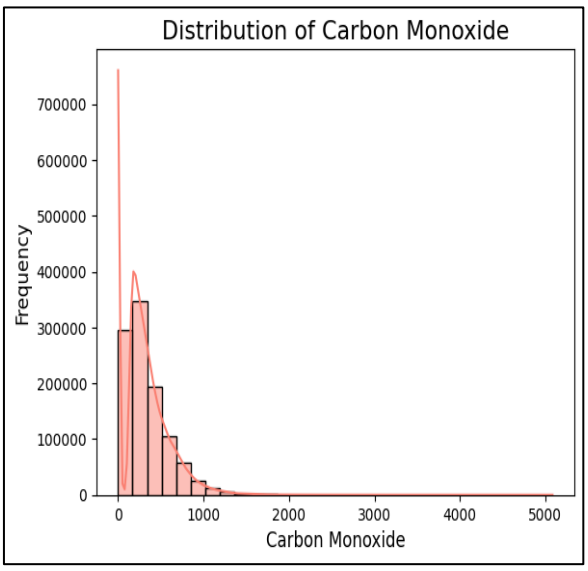


Fig 5: Histogram of CO.

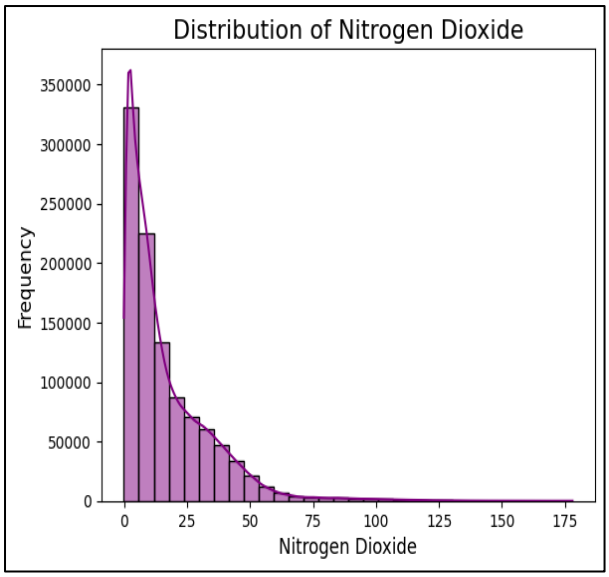


Fig 6: Histogram of NO_2..

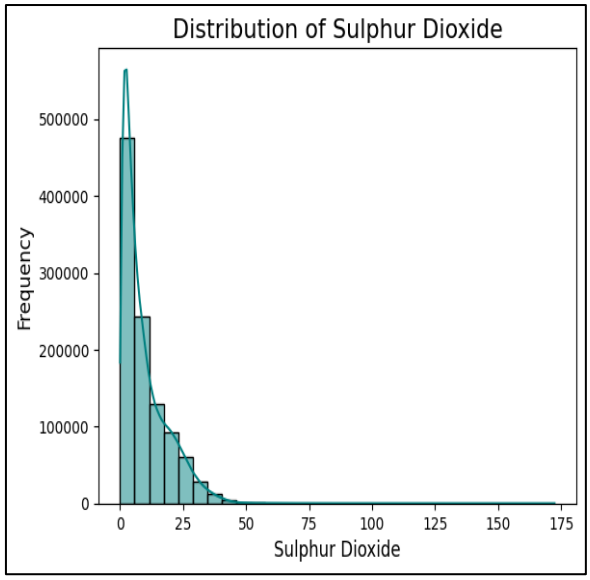


Fig 7: Histogram of SO_2.

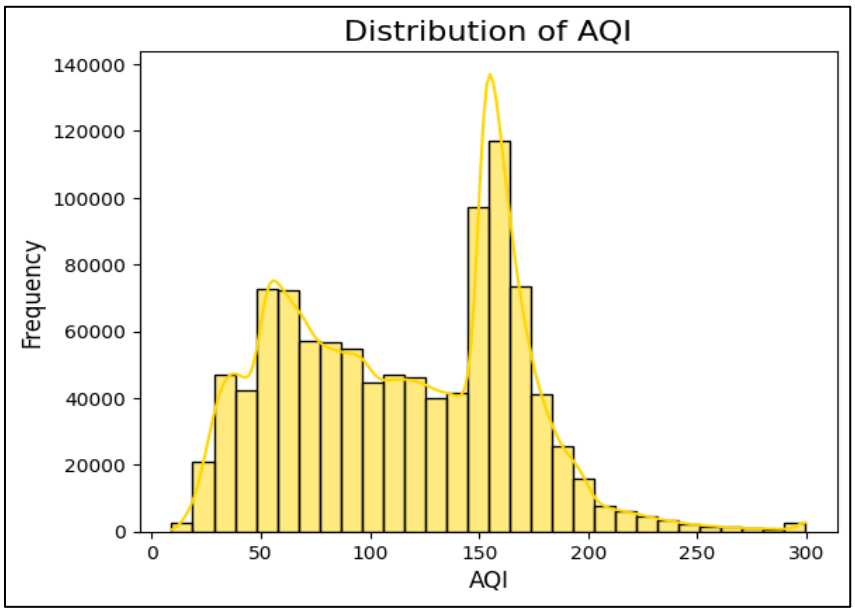


Fig 8: Histogram of AQI

The histogram analysis shows that the majority of pollutant variables are positively skewed, with a high concentration of lower values and tail toward higher value. Ozone has a more symmetric distribution, whereas AQI has a multimodal pattern that reflects changing air quality conditions over time and in different areas. Overall the data indicates deviation and presence of extreme values.

This figure shows the distribution of major pollutants. PM2.5 and PM10 are heavily right-skewed, indicating frequent pollution spikes in urban regions.

3.3 Boxplots and Outlier Analysis

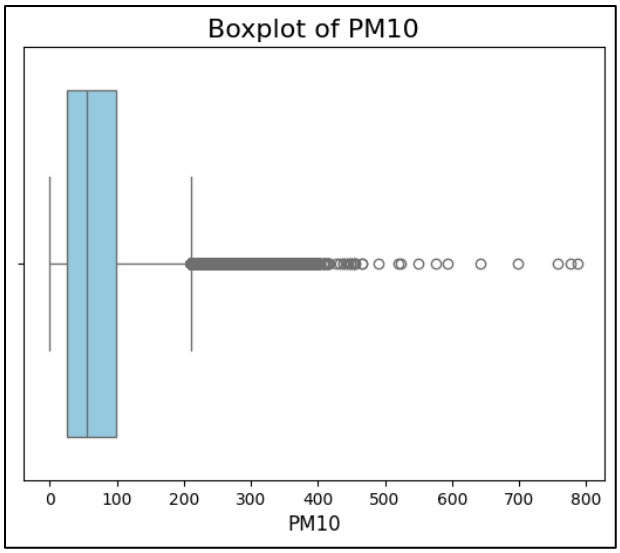


Fig 9: BoxPlot of PM10.

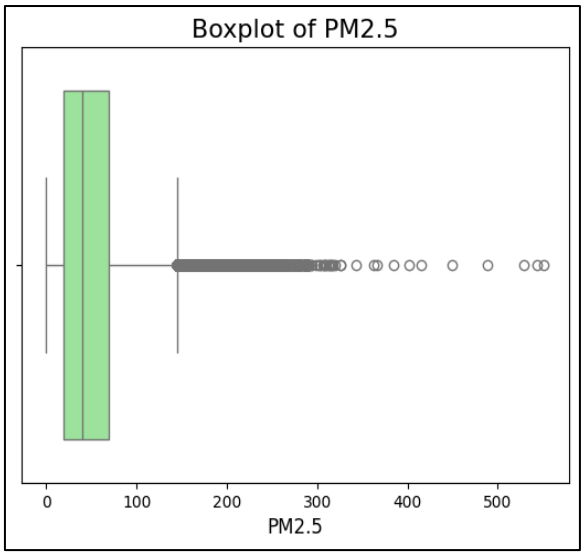


Fig 10: BoxPlot of PM2.5.

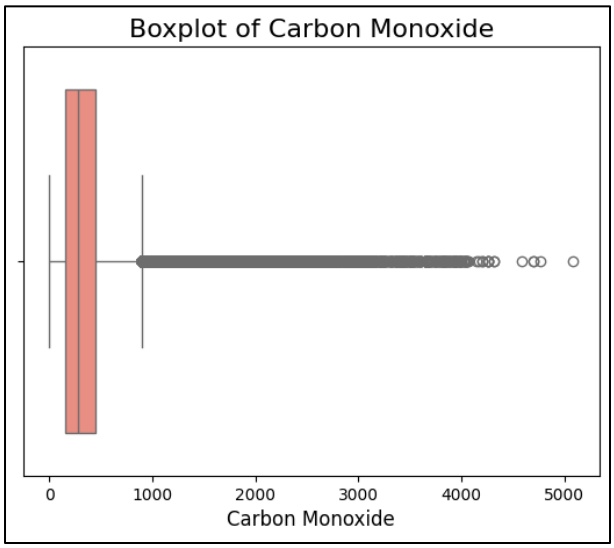


Fig 11: BoxPlot of CO.

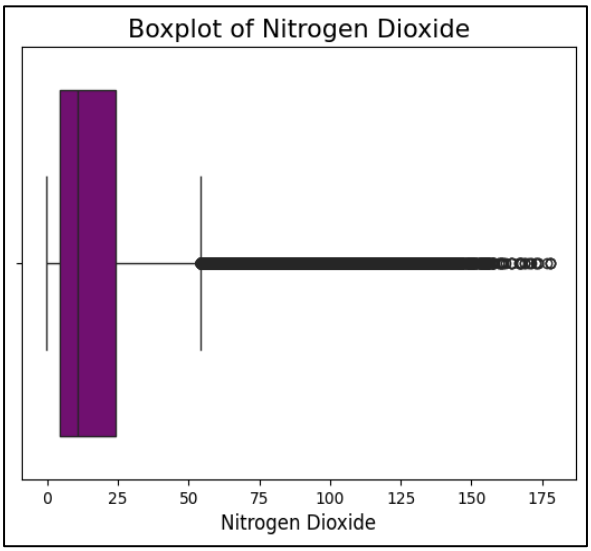


Fig 12: BoxPlot of NO_2.

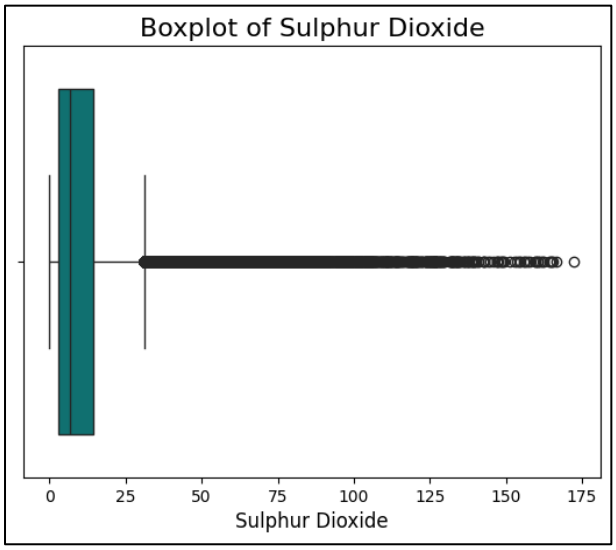


Fig 13: BoxPlot of SO_2.

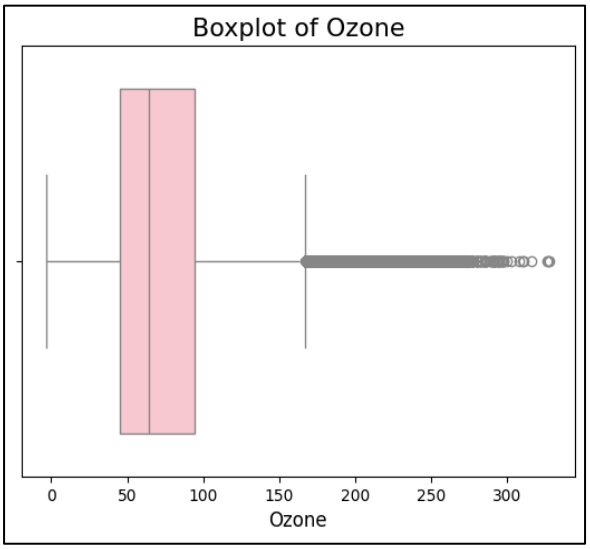


Fig 14: BoxPlot of O_3.

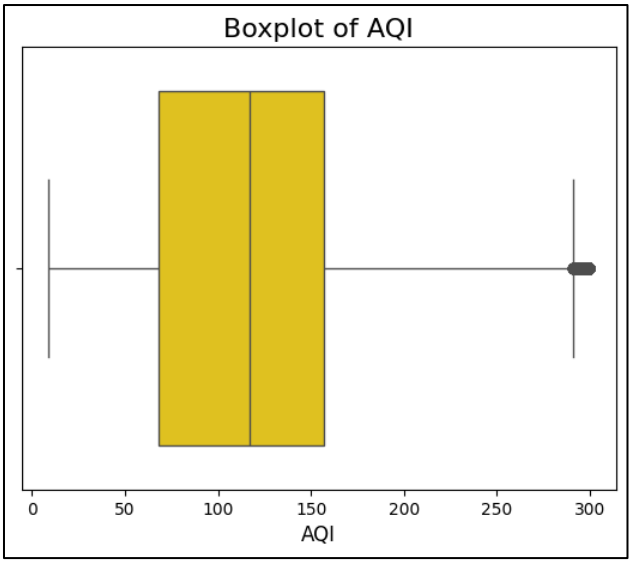


Fig 15: BoxPlot of AQI.

Feature	Outliers count (Using IQR Rules)
PM10	56457
PM2.5	40375
Carbon monoxide	38983
Nitrogen dioxide	35237
Sulphur dioxide	32759
Ozone	34935
AQI	2630

TABLE IV: Outliers count of each feature.

The boxplots demonstrate that all air quality indicators are right-skewed, with many outliers indicating that pollution levels can be higher on occasion. PM10 and PM2.5 show the most variability and outliers, indicating that certain concentrations of air pollutants are the primary cause of air quality variations. Overall, it shows unstable situations with occasional pollution increase.

3.4 Temporal Analysis

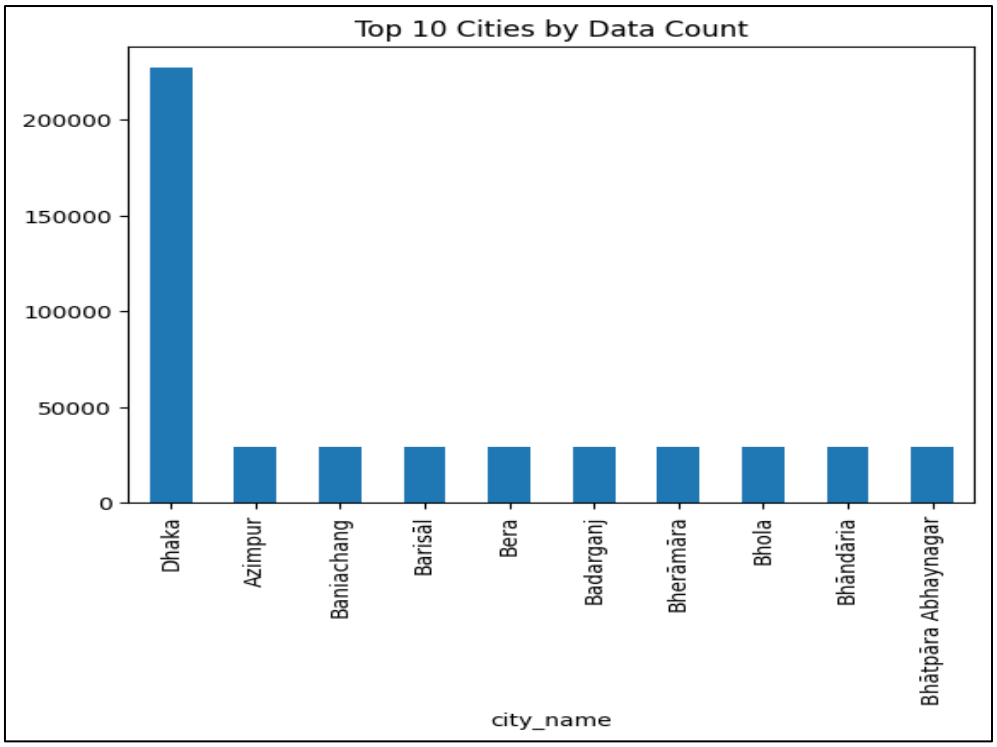


Fig 16: Bar Chart of Top 10 Cities by data count

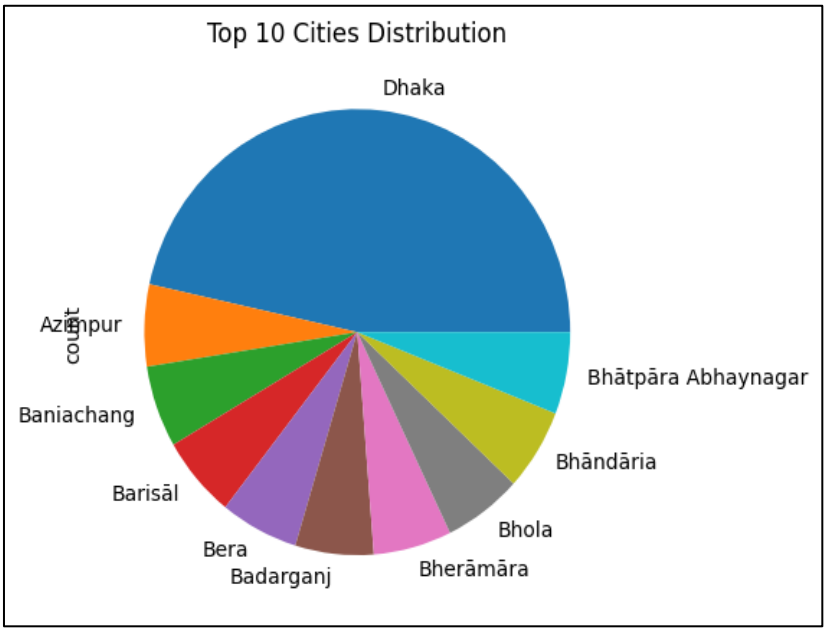


Fig 17: Pie Chart of Top 10 Cities by data count

The most frequent category is Dhaka acquiring 22% of total data, which is a much larger amount of data than the other cities. The remaining cities (Azimpur, Baniachang, Barisal, and others) have comparable but considerably lower counts. This suggests a significant data imbalance, with Dhaka heavily overrepresented in comparison to other areas, which may bias the analysis towards Dhaka's conditions.

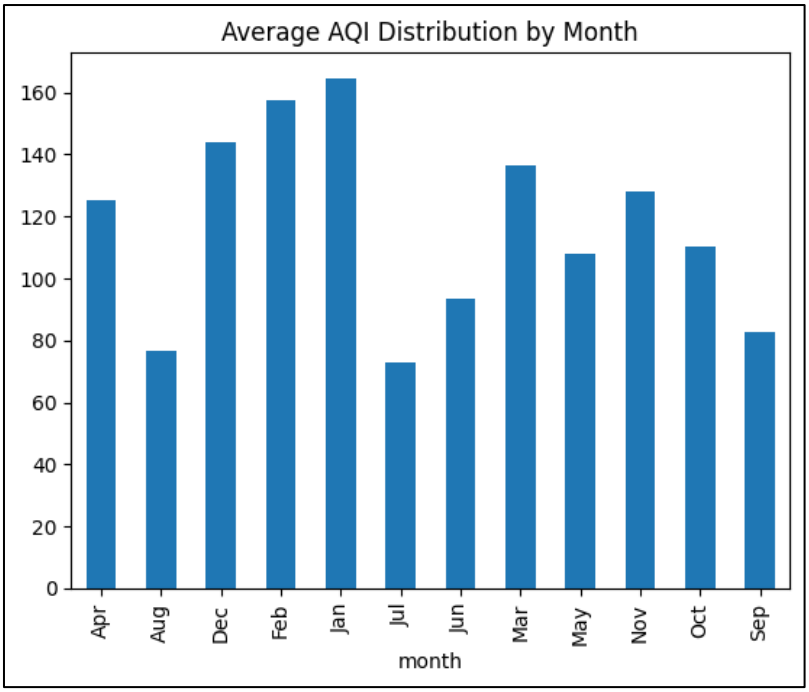


Fig 18: Average AQI distribution by month

The analysis reveals a clear seasonal variation in air quality, with the highest pollution levels occurring during winter months (January–February) and the lowest during monsoon months (July–August). This suggests that meteorological factors such as rainfall and temperature play a significant role in influencing air pollution levels.

3.5 Scatterplots & Correlation Heatmap

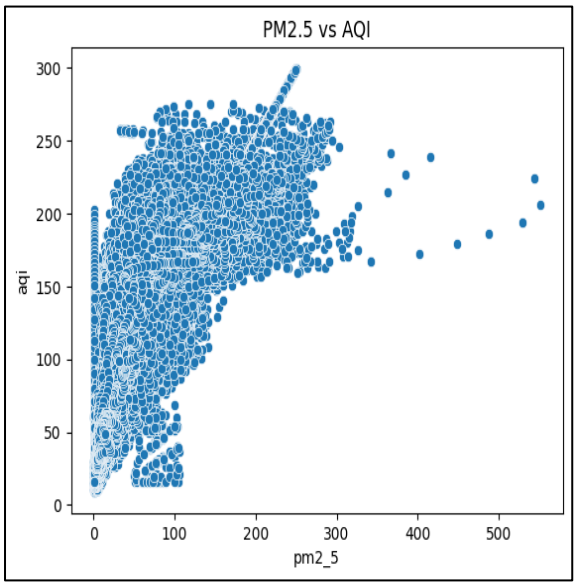


Fig 19: Scatterplot of PM2.5 vs AQI.

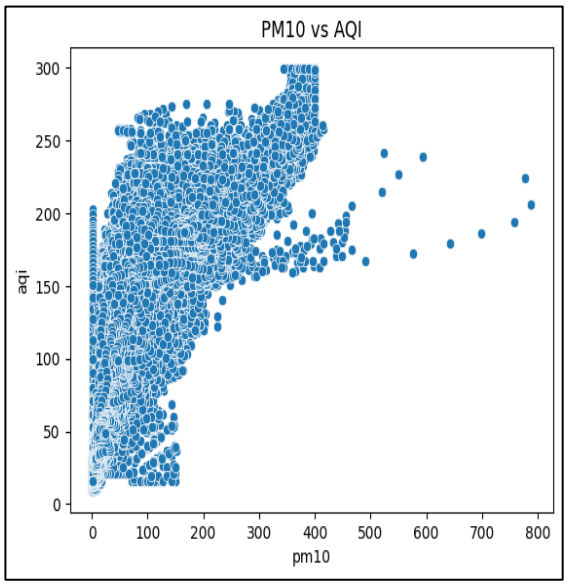


Fig 20: Scatterplots of PM10 vs AQI.

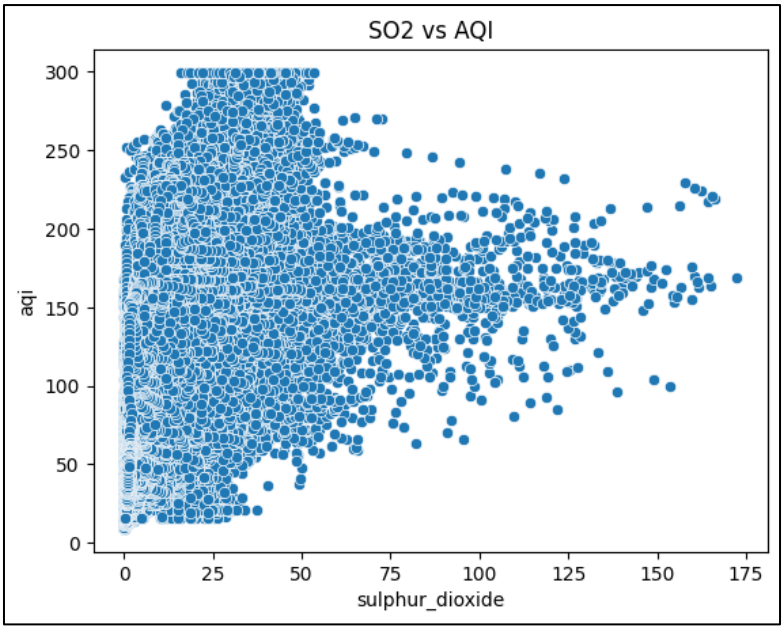


Fig 21: Scatterplots of SO_2 vs AQI

The scatter plots indicate that the contaminants and AQI have a positive correlation.PM2.5 and AQI (0.87) and PM10 and AQI (0.84) show strong positive correlations, which are the reason for the evident upward trend in their plots. Carbon monoxide and AQI (0.19) have a moderately positive correlation, which indicates a more scattered but still increasing pattern. Overall, PM2.5 and PM10 have the greatest impact on AQI, with others having lower to moderate effects.

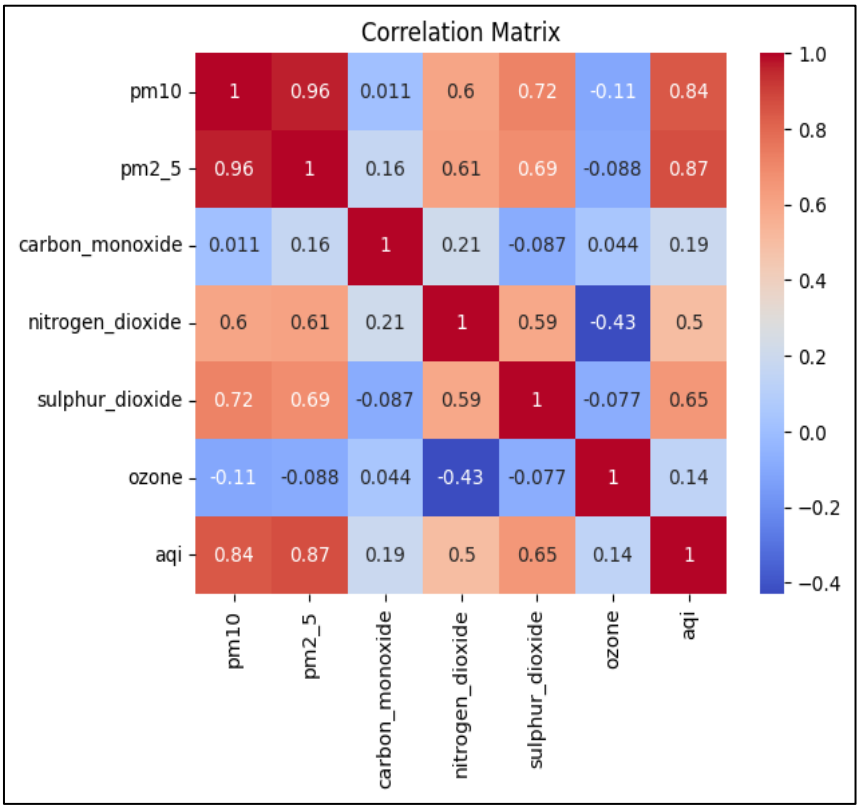


Fig 22: Correlation Matrix of PM10, PM2.5, CO, NO_2,SO_2, O_3, AQI

The Correlation Heatmap shows that PM2.5 and PM10 have the strongest positive correlation with AQI. This indicates that particulate matter is the primary contributor to air pollution levels in Bangladesh. These variables are the most influential in determining air quality. A very high correlation between PM2.5 and PM10 suggests that these pollutants often increase or decrease together, likely due to common sources such as vehicle emissions, construction activities, and industrial pollution.

Pollutants like nitrogen dioxide (NO_2) and sulphur dioxide (SO_2) show moderate correlation with AQI, indicating that they contribute to air pollution but are less dominant compared to particulate matter.

Ozone (O_3) and carbon monoxide (CO) exhibit weak correlations with AQI, suggesting that their impact on overall air quality variation is relatively low in this dataset.

4. Methodology

This study applies both linear and non-linear machine learning techniques to predict Air Quality Index (AQI) using pollutant data. The methodology includes two main modeling approaches: Linear Regression and Random Forest Regression.

4.1 Linear Regression

Linear Regression was used as a baseline model to understand the relationship between pollutant concentrations and AQI.
The model assumes a linear relationship between input features and the target variable:

$$\hat{y} = m_1 x_1 + m_2 x_2 + \cdots + b \qquad \text{----------(1)}$$

Where:

- x_i represent pollutant concentrations
- m_i are model coefficients
- b is the intercept

The dataset was divided into:

- **70% training data**
- **30% testing data**

This split ensures that the model is trained on a majority of the data while being evaluated on unseen samples for generalization.

Two feature sets were tested:

- **Model 1:** PM2.5, PM10
- **Model 2:** PM2.5, PM10, SO_2

4.1.1 Model 1 (PM2.5 and PM10)

The first Linear Regression model was trained using PM2.5 and PM10 as input features.

Learned Equation:

$$AQI = 59.54 + 1.09 \cdot PM2.5 + 0.00 \cdot PM10 \quad \text{----------(2)}$$

Performance (Test Set):

- MSE: 668.68
- RMSE: 25.86
- MAE: 21.09
- R^2: 0.7589

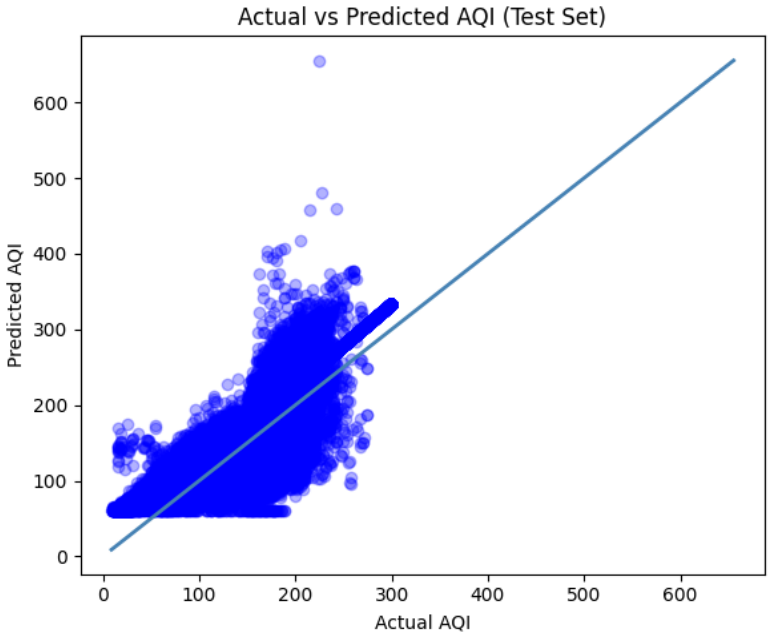


Figure 23: Actual vs Predicted AQI (Linear Regression Model 1)

This figure compares the predicted AQI values with the actual AQI values for Model 1. Ideally, all points should lie on the diagonal line. However, noticeable deviations occur, particularly at higher AQI values, indicating that the model struggles to accurately predict extreme pollution conditions.

PM2.5 has a strong positive impact on AQI. PM10 coefficient is nearly zero, suggesting minimal contribution in this model. The model captures general trends but lacks precision for higher AQI values.

4.1.2 Model 2 (PM2.5, PM10 & SO_2)

To improve performance, SO2 was added as an additional feature.

Learned Equation:

$$AQI = 57.90 + 1.10 \cdot PM2.5 - 0.07 \cdot PM10 + 0.66 \cdot SO2 \quad \text{------(3)}$$

Performance (Test Set):

- MSE: 650.14
- RMSE: 25.50
- MAE: 20.67
- R^2: 0.765

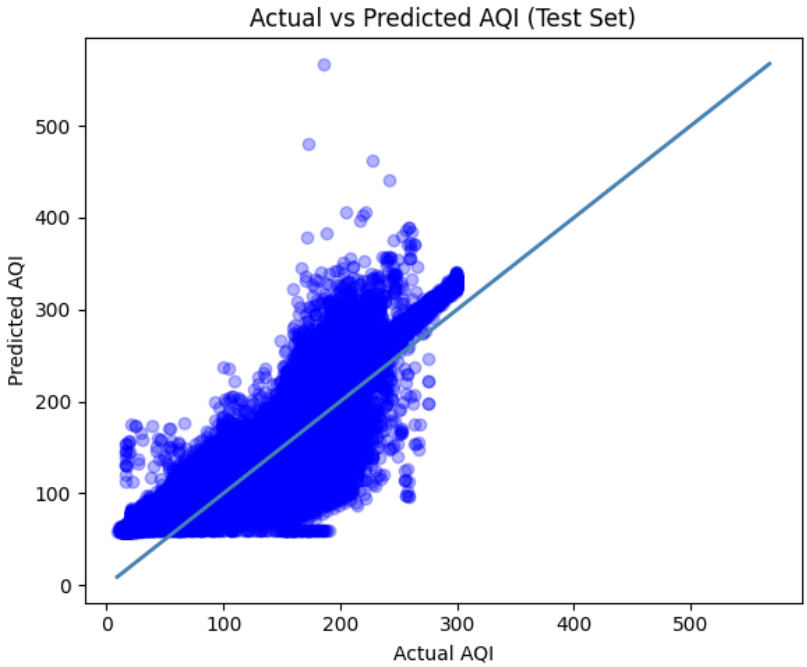


Figure 24: Actual vs Predicted AQI (Linear Regression Model 2)

This figure shows improved alignment between predicted and actual AQI values compared to Model 1. The inclusion of SO2 enhances the model's ability to capture variations in AQI, although some deviation still exists for extreme values.

PM2.5 remains the dominant predictor. SO2 contributes positively to AQI prediction. PM10 shows a small negative coefficient, indicating a weaker or inverse relationship in the presence of other variables. Overall performance improved compared to Model 1.

4.2 Random Forest

Random Forest Regression was applied as a non-linear ensemble learning method to predict AQI. This model is capable of capturing complex relationships between pollutant concentrations and AQI without assuming linearity.

The dataset was divided into:

- **70% training data**
- **30% testing data**

To enhance model performance, the following preprocessing steps were applied:

1. Extracted temporal features from datetime:
    - **Hour**
    - **Day**
    - **Month**
2. Removed non-informative features:
    - **city_id**
    - **city_name**

The Random Forest model was configured with 200 decision trees and a maximum tree depth of 15 to balance model complexity and generalization performance.

4.2.1 Model 1: Full Feature Set (Best Model)

Features:

lat, lon, pm10, pm2_5, carbon_monoxide, nitrogen_dioxide, sulphur_dioxide, ozone, hour, day, month.

Performance (Test Set):

- MAE: 8.27
- MSE: 159.61
- RMSE: 12.63
- R^2: 0.94

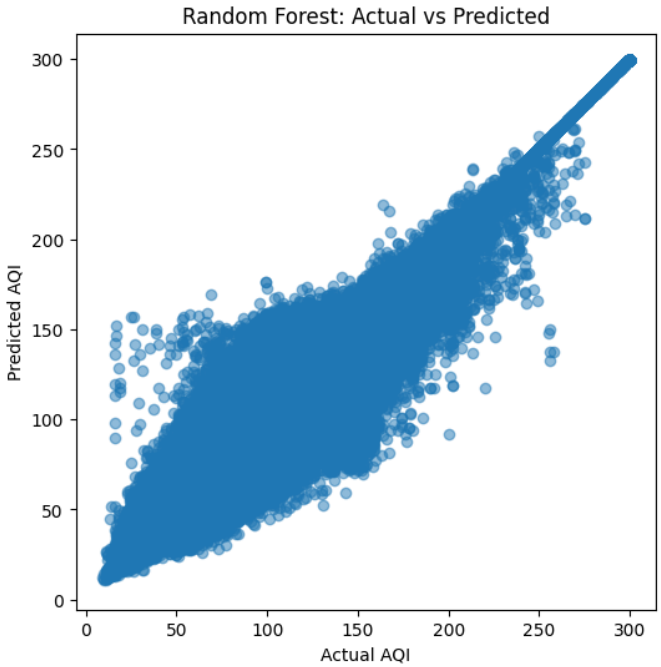


Figure 25: Actual vs Predicted AQI (Random Forest Model 1)

The Actual vs Predicted plot shows a strong linear alignment, indicating that the Random Forest model captures the relationship between pollutant concentrations and AQI effectively. However, increased dispersion at higher AQI levels suggests reduced accuracy under extreme pollution conditions, with a tendency to slightly underestimate peak AQI values.

pm2_5	0.878310
pm10	0.032115
ozone	0.029546
carbon_monoxide	0.019543
month	0.010778
hour	0.007338
day	0.006821
lon	0.005259
lat	0.003501
nitrogen_dioxide	0.003462
sulphur_dioxide	0.003326

Figure 26: Feature Importance (Model 1).

PM2.5 dominates the prediction with an importance of ~0.878, confirming it as the most critical factor in AQI determination. Other features such as PM10, ozone, and carbon monoxide contribute marginally, while spatial and temporal features have relatively small influence.

4.2.2 Model 2: Pollutants Only

Features:
pm10, pm2_5, carbon_monoxide, nitrogen_dioxide, sulphur_dioxide, ozone.

Performance (Test Set):

- MAE: 10.04
- MSE: 231.71
- RMSE: 15.22
- R^2: 0.92

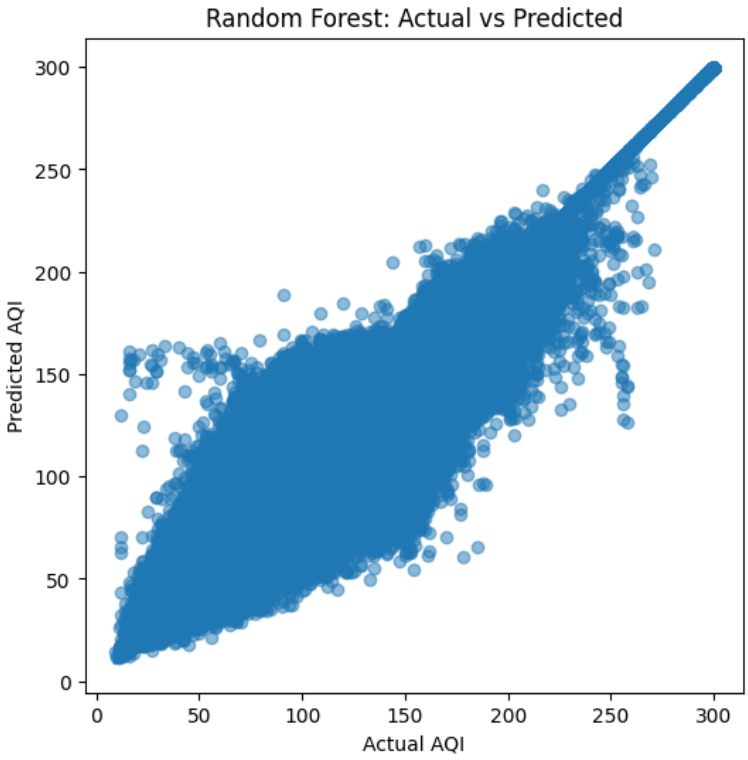


Figure 27: Actual vs Predicted AQI (Random Forest Model 2)

This plot shows a strong linear alignment, indicating that the Random Forest model captures the relationship between pollutant concentrations and AQI effectively. However, increased dispersion at higher AQI levels suggests reduced accuracy under extreme pollution conditions, with a tendency to slightly underestimate peak AQI values.

pm2_5	0.897044
pm10	0.034532
ozone	0.033018
carbon_monoxide	0.021040
nitrogen_dioxide	0.008489
sulphur_dioxide	0.005876

Figure 28: Feature Importance (Model 2).

PM2.5 again dominates the prediction with an importance of ~0.897, confirming it as the most critical factor in AQI determination. Other features such as PM10, ozone, and carbon monoxide contribute marginally, while spatial and temporal features have relatively small influence.

4.2.3 Model 3: PM2.5 and PM10 Only

Features:

pm10, pm2_5.

Performance (Test Set):

- MAE: 13.25
- MSE: 383.37
- RMSE: 19.58
- R^2: 0.85

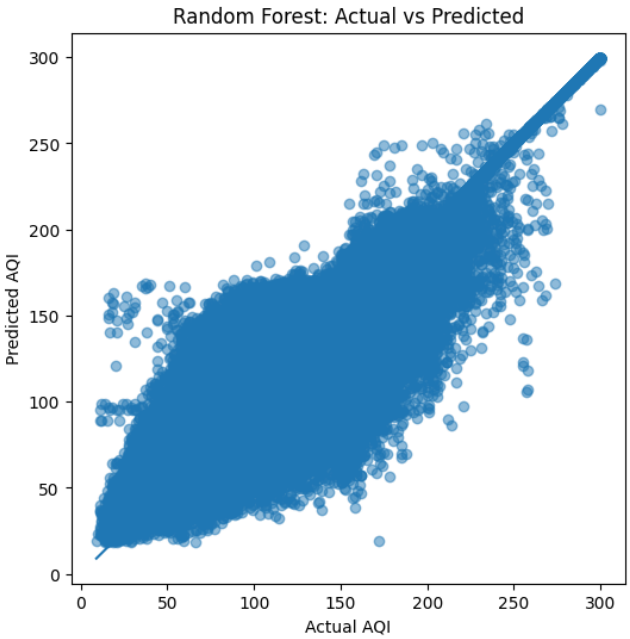


Figure 29: Actual vs Predicted AQI (Random Forest Model 3)

The Actual vs Predicted plot shows almost same as Model1 & Model2 with strong linear alignment, indicating that the Random Forest model captures the relationship between pollutant concentrations and AQI effectively.

```
pm2_5     0.965091
pm10      0.034909
```

Figure 30: Feature Importance (Model 3).

Again PM2.5 dominates the prediction with an importance of ~0.96, confirming it as the most critical factor in AQI determination. Other features such as PM10, ozone, and carbon monoxide contribute marginally, while spatial and temporal features have relatively small influence.

4.3 Methodological Framework

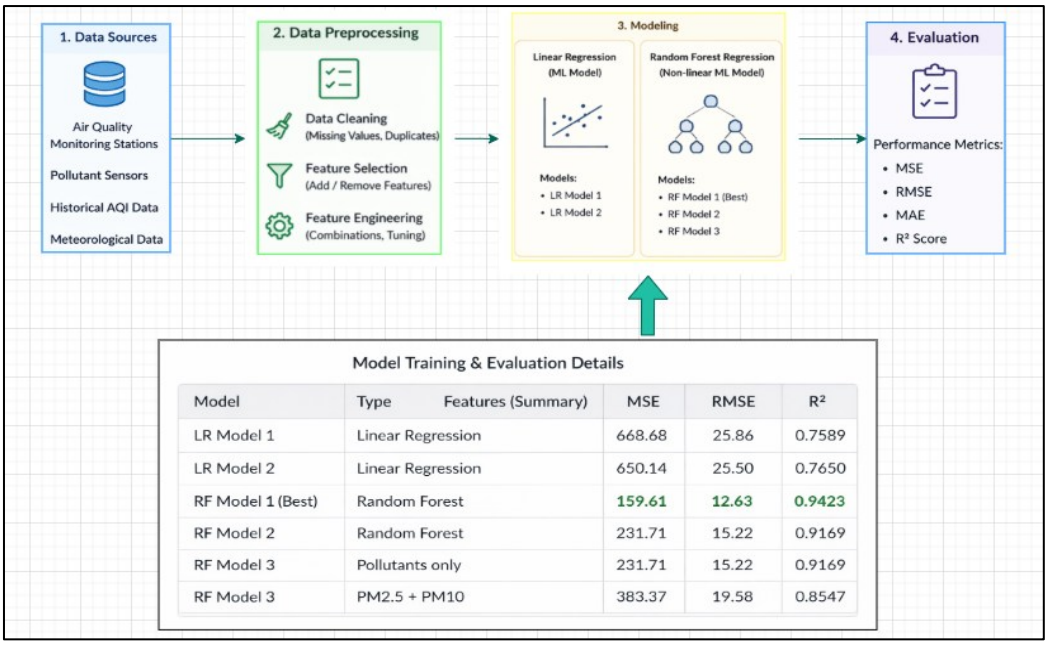


Figure 31: Methodological Framework of whole process.

5. Performance Comparison of All Models

Model	**Type**	**Features (Summary)**	**MSE**	**RMSE**	**MAE**	**R²**
LR Model 1	Linear Regression	PM2.5, PM10	668.68	25.86	21.09	0.7589
LR Model 2	Linear Regression	PM2.5, PM10, SO2	650.14	25.50	20.67	0.765
RF Model 1 (Best)	Random Forest	All features + tuned	159.61	12.63	8.27	0.9423
RF Model 2	Random Forest	Pollutants only	231.71	15.22	10.04	0.9169
RF Model 3	Random Forest	PM2.5 + PM10	383.37	19.58	13.25	0.8547

TABLE V: Performance comparison of all models.

The large performance gap between Linear Regression and Random Forest confirms that AQI is influenced by complex, non-linear interactions among pollutants and environmental factors.

6. Conclusion

This study demonstrates that machine learning models can effectively predict AQI using pollutant and temporal data.

Random Forest Regression achieved the best performance due to its ability to capture non-linear relationships, while Linear Regression provided a simple and interpretable baseline.

The results highlight the importance of particulate matter (PM2.5 and PM10) as primary drivers of air pollution in Bangladesh.

7. Future Work

Future improvements of this study can focus on developing a real-time AQI prediction system data and deploying the trained model into a web or mobile application for continuous air quality monitoring and early warning.

Additionally, advanced deep learning models such as LSTM (Long Short-Term Memory) can be applied to capture temporal dependencies in air pollution data. Since AQI is highly time-dependent, LSTM-based models can improve forecasting accuracy by learning long-term patterns in pollutant variations.

Reference

1. Paul, Tapon (2026), "Bangladesh Air Quality Index (AQI) Dataset (2000–2025): Historical Hourly Air Pollution Data Across 103 Cities", Mendeley Data, V2, doi: 10.17632/9j447cynb9.2

Important Links

- LR (Model 1) Code: https://www.kaggle.com/code/sakinofbd/cse303-linear-regression-pm2-5-pm10
- LR (Model 2) Code: https://www.kaggle.com/code/sakinofbd/cse303-linear-regression-pm2-5-pm10-so2
- RF (Model 1) Code: https://www.kaggle.com/code/sakinofbd/cse303-randomforest-all-features
- RF (Model 2) Code: https://www.kaggle.com/code/sakinofbd/cse303-randomforest-pm10-pm2-5-co-no2-so2-o3
- RF (Model 3) Code: https://www.kaggle.com/code/sakinofbd/cse303-randomforest-pm10-pm2-5
- Project Initial Todos: https://www.kaggle.com/code/sakinofbd/cse303-project-initial-todos-data-overthinkers

The End